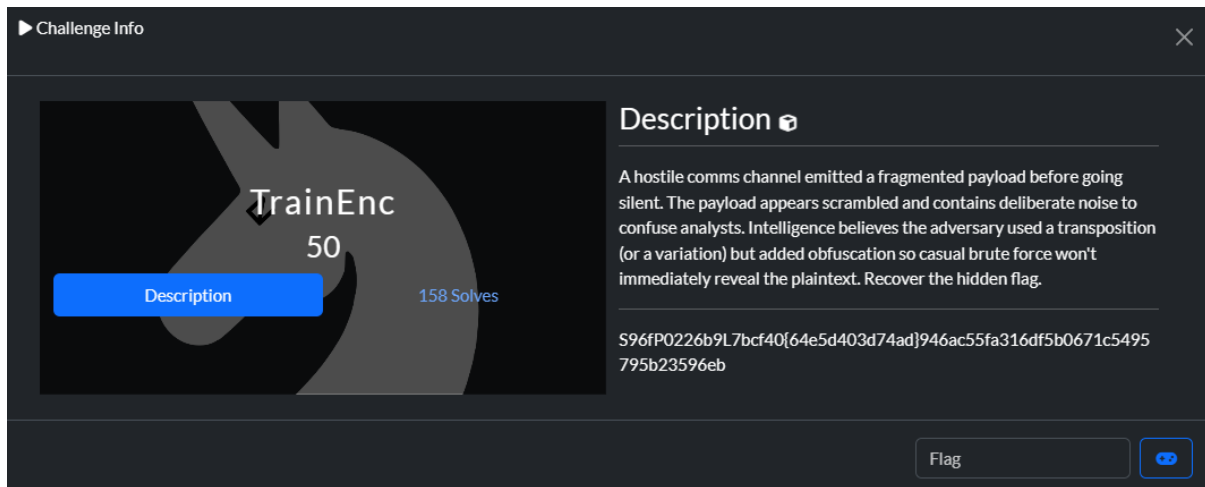


Challenge Name: TrainEnc - 50 (CRYPTO)



Challenge Info

TrainEnc 50 158 Solves

Description

A hostile comms channel emitted a fragmented payload before going silent. The payload appears scrambled and contains deliberate noise to confuse analysts. Intelligence believes the adversary used a transposition (or a variation) but added obfuscation so casual brute force won't immediately reveal the plaintext. Recover the hidden flag.

S96fP0226b9L7bcf40{64e5d403d74ad}946ac55fa316df5b0671c5495795b23596eb

Flag

The challenge provided the following scrambled payload:

```
S96fP0226b9L7bcf40{64e5d403d74ad}946ac55fa316df5b0671c5495795b23596eb
```

According to the description, the payload:

- Was **fragmented and scrambled**.
- Contained **deliberate noise** to confuse analysts.
- Likely used a **transposition cipher or variation** with added obfuscation.

Step 1 – Inspect and clean the payload:

I first observed the payload and removed obvious **non-hex noise** (like letters **S**, **P**, **L** and braces outside the hex). This left a mostly hex-formatted string suitable for further analysis.

Step 2 – Detect transposition:

The challenge hinted at **columnar or similar transposition**, meaning the hex digits were rearranged. Since the adversary added deliberate noise, a simple linear read wouldn't reveal the plaintext.

Step 3 – Brute-force likely widths:

I performed a brute-force over different column widths to reverse the transposition. This approach revealed the original 64-byte hex token.

Step 4 – Reconstruct the final flag:

Using the header markers **S**, **P**, **L** as guidance, I formatted the recovered token into the expected flag format:

```
SPL{09a0597634367092b4d61796516a7ec266f54cdcbe25f5ad4bf904d5543b59bf}
```

Conclusion:

- The challenge involved **noise removal, hex inspection, and reversing a columnar transposition.**
- The adversary tried to obscure the payload with deliberate letters and scrambled hex order, but systematic brute-force over transposition widths allowed recovery of the hidden flag.

Flag: `SPL{09a0597634367092b4d61796516a7ec266f54cdcbe25f5ad4bf904d5543b59bf}`

`~By: 1nf1n1ty`